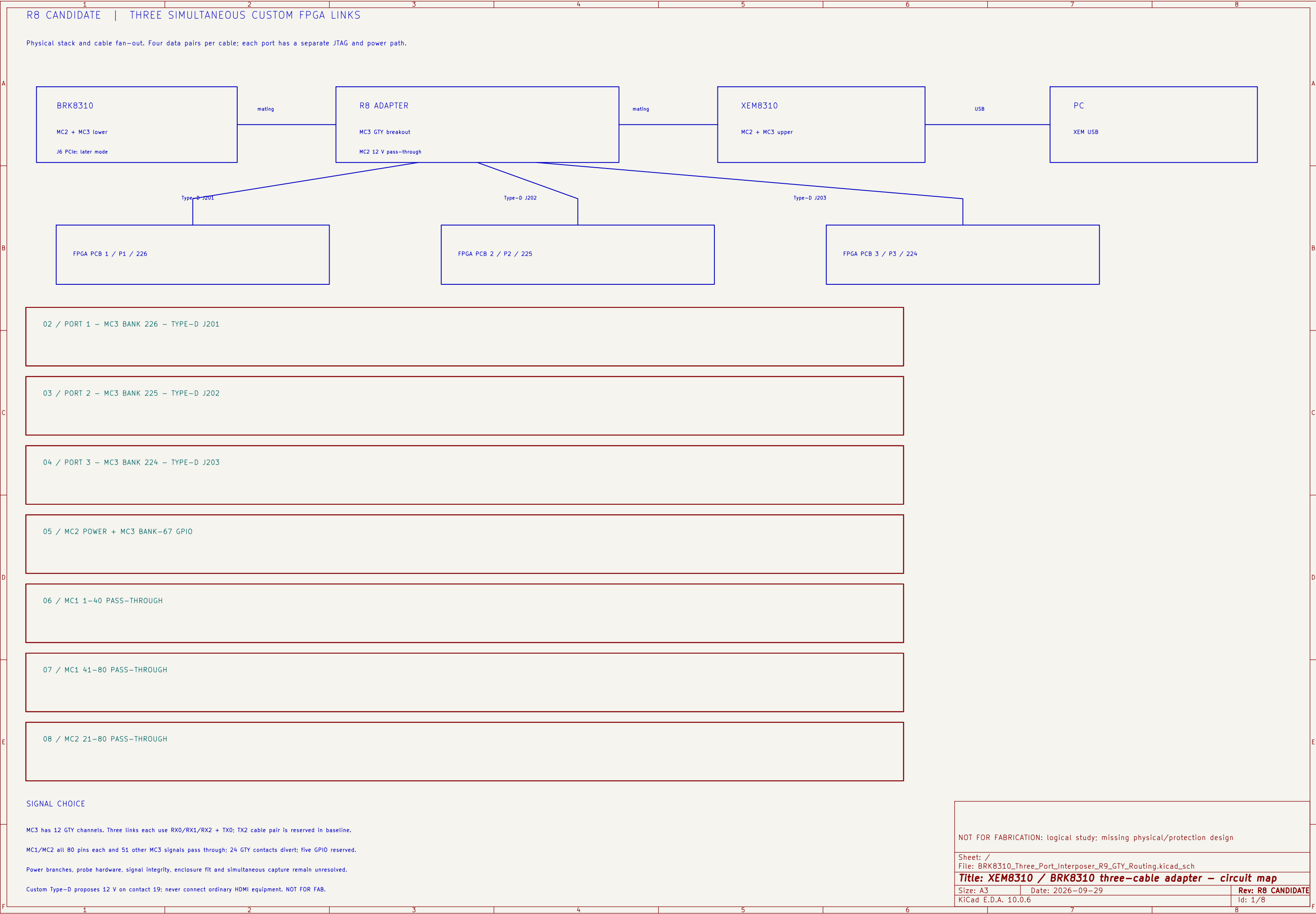
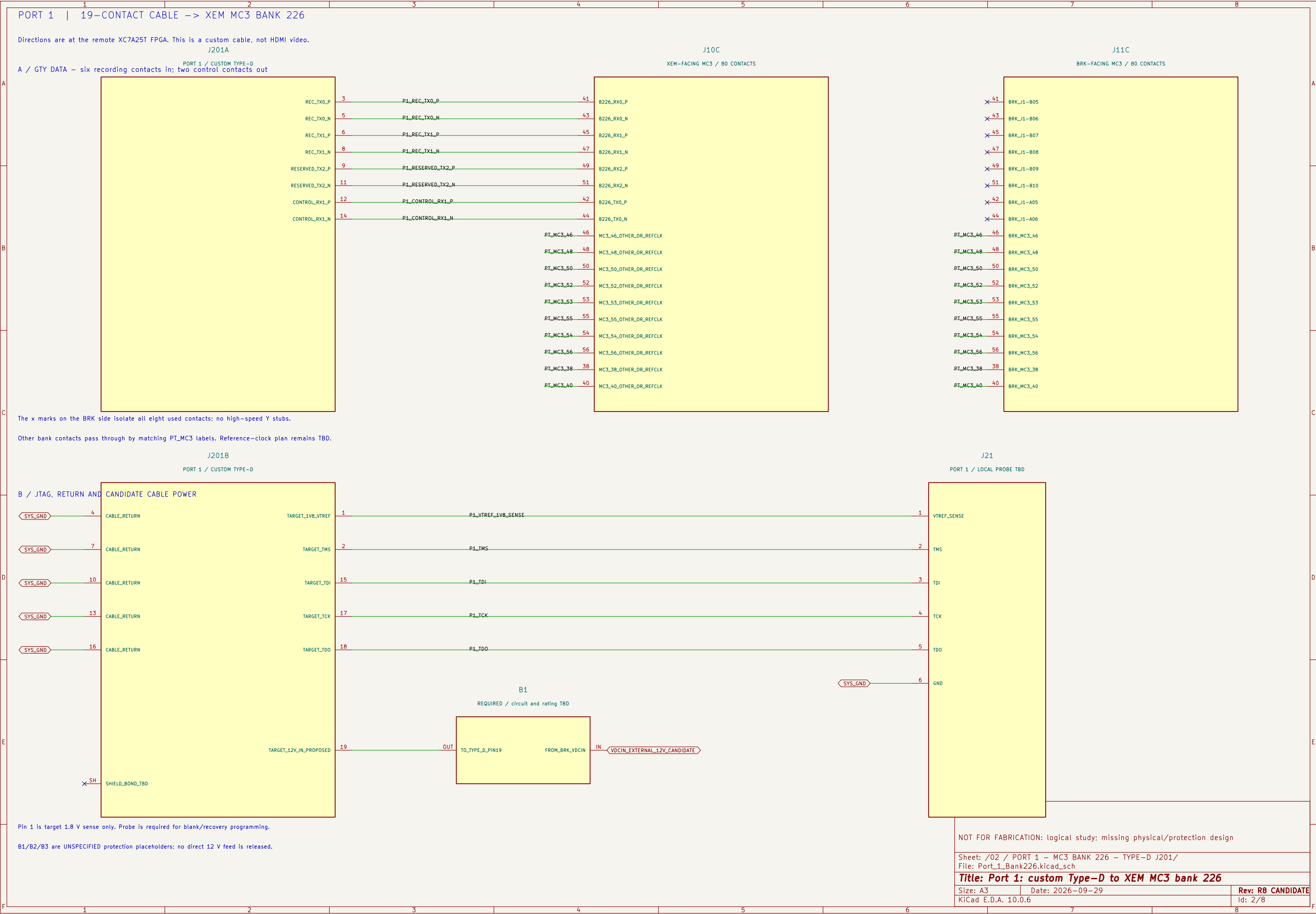
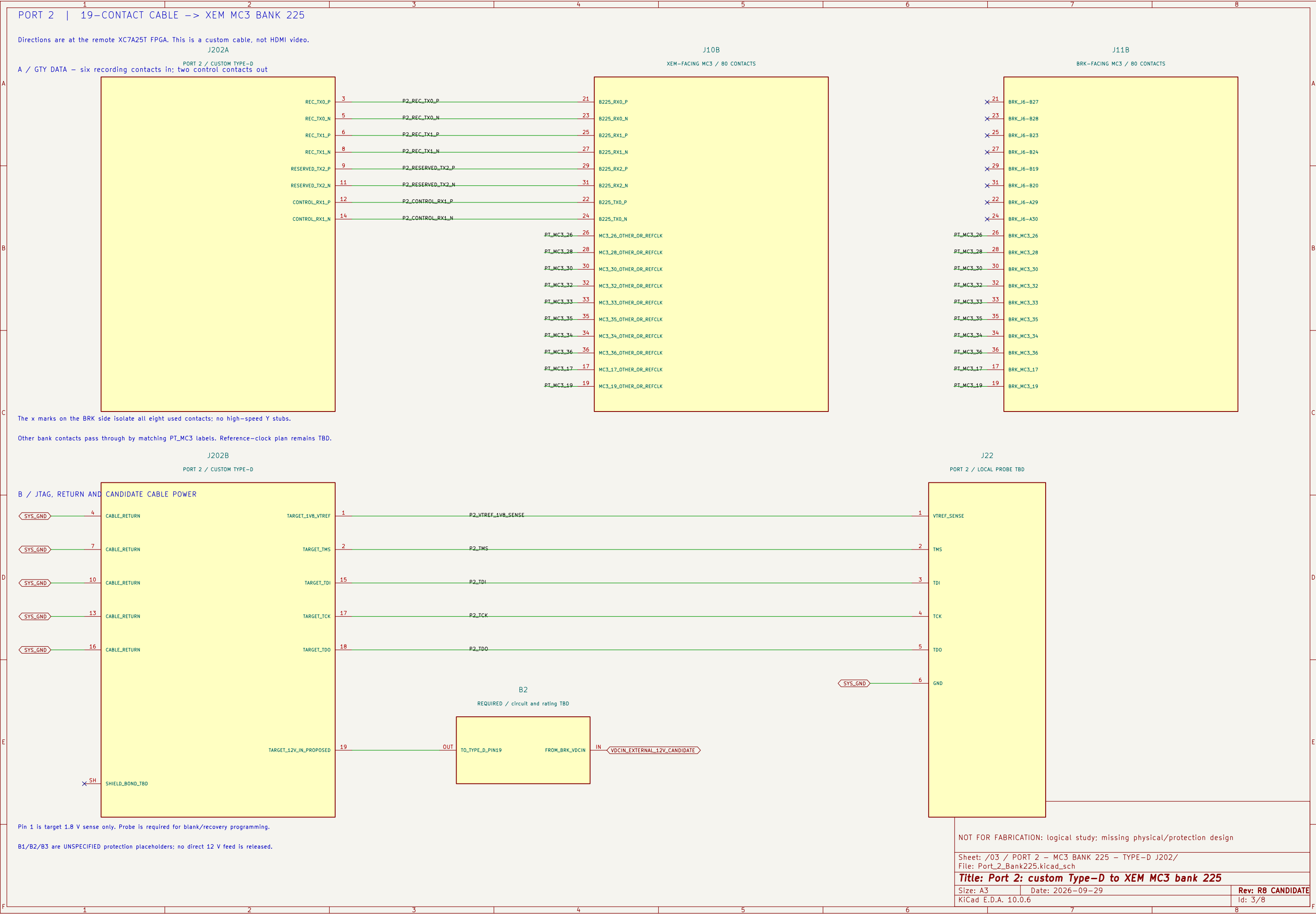
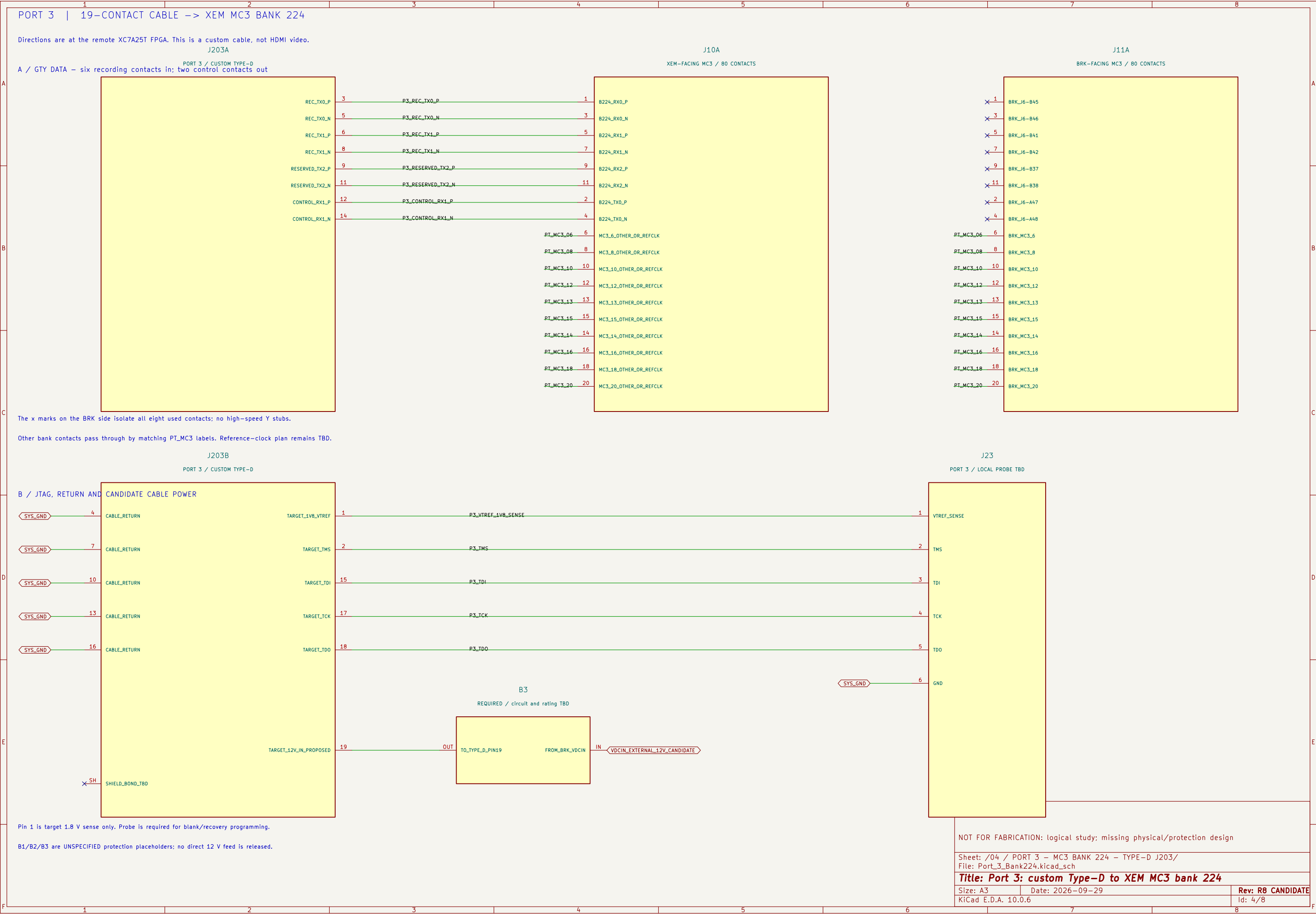






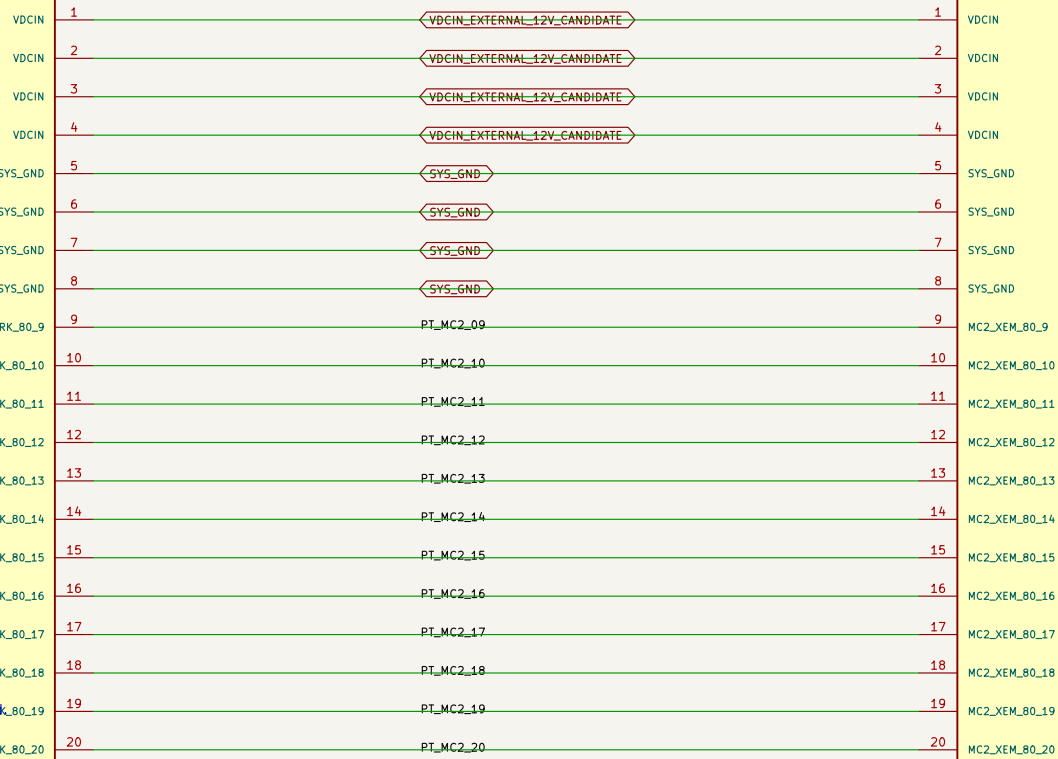




MC2 POWER | EXTERNAL 12 V THROUGH BRK TO XEM

The XEM does not generate 12 V. Its VDCIN contacts receive the external BRK supply.

MC2 contacts 1-4 = VDCIN; 5-8 = DGND. Contacts 9-20 pass through here too.

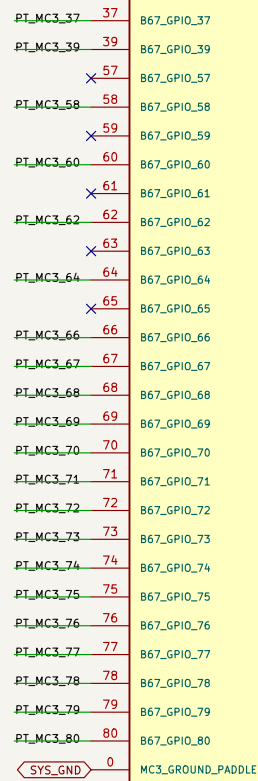


BRK power input and the three 12 V branch loads require a common current/fault budget.

Separate branch protection, cable/contact current, inrush and remote regulators: unregulated

MC1 1-80 and MC2 21-80 continue on sheets 06-08; all are in the native netlist.

MC3 BANK-67 GPIO | BRIDGE RESERVATION



57/59/61/63/65: isolated reservations for a possible XEM JTAG bridge.

Bank 67 is 1.2 V. No translator, source select or gateway is implemented.

NOT FOR FABRICATION: logical study; missing physical/protection design

Sheet: /05 / MC2 POWER + MC3 BANK-67 GPIO/
File: MC2_Power_and_MC3_GPIO.kicad_sch

Title: MC2 input power and MC3 bank-67 GPIO

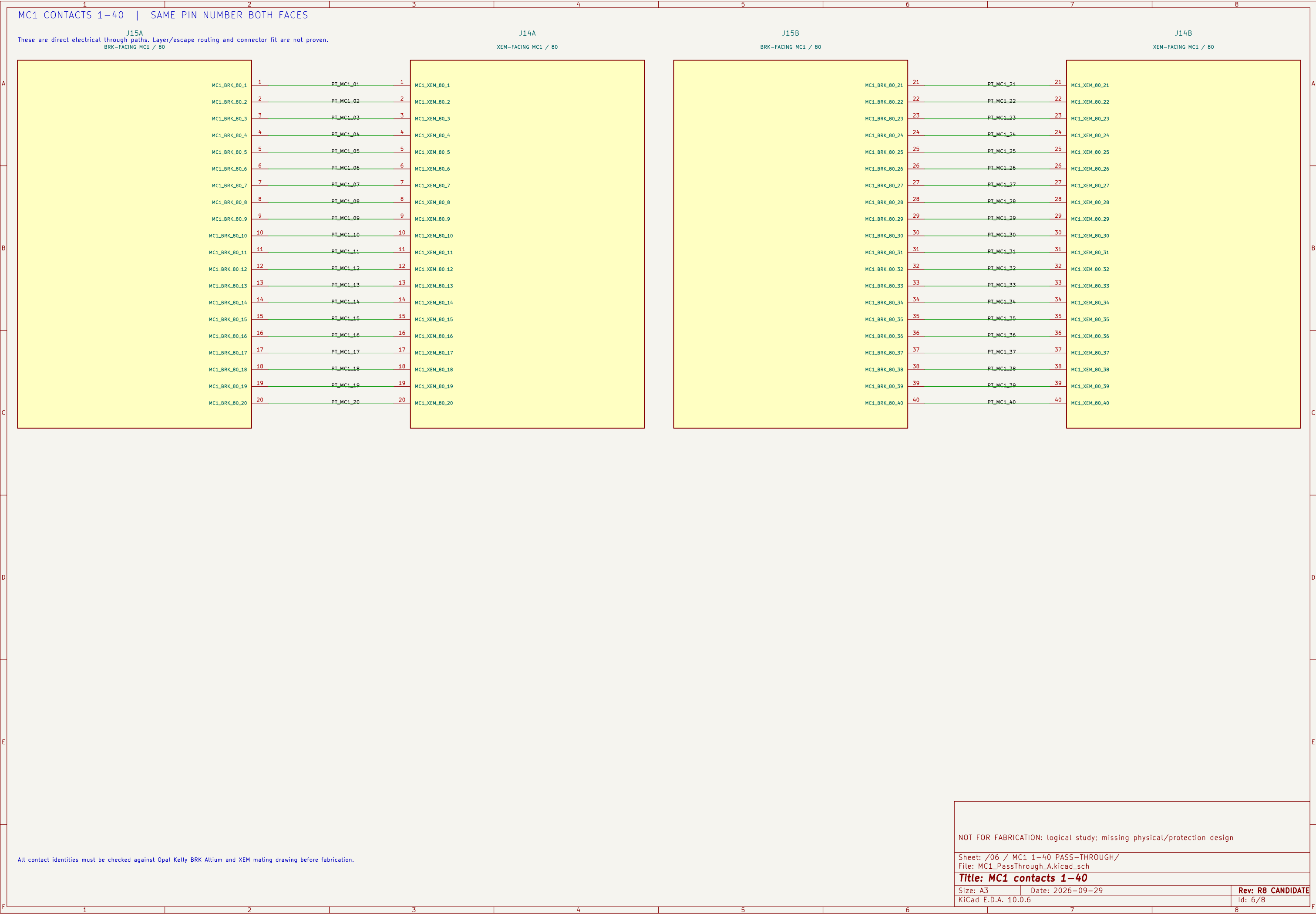
Size: A3

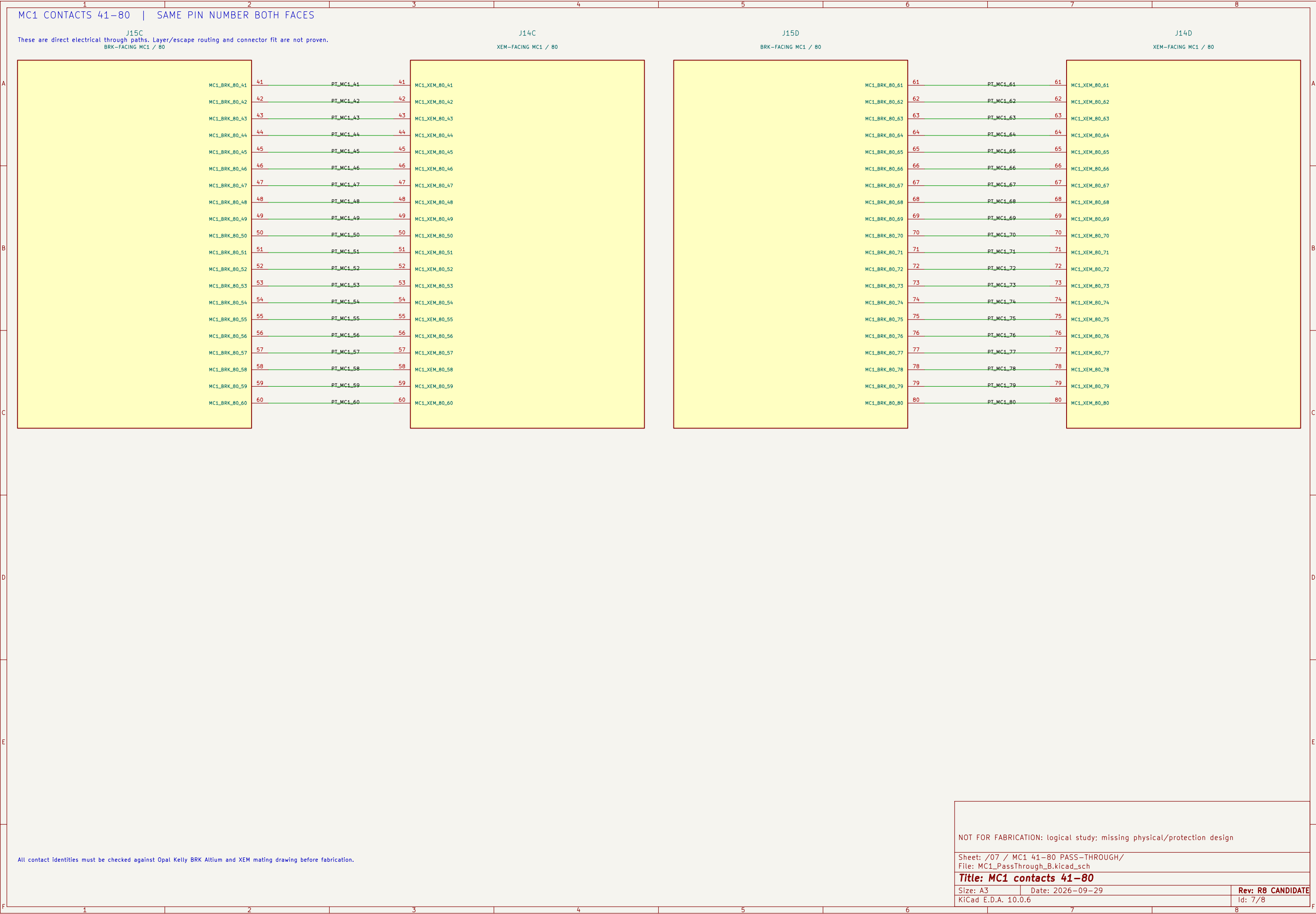
Date: 2026-09-29

Rev: R8 CANDIDATE

KiCad E.D.A. 10.0.6

0.6 Id: 5/8





	1	2	3	4	5	6	7	8		
	MC2 CONTACTS 21–80 SAME PIN NUMBER BOTH FACES									
	J13B These are direct electrical through paths. Layer/escape routing and connector fit are not proven. BRK–FACING MC2 / 80		J12B XEM–FACING MC2 / 80		J13C BRK–FACING MC2 / 80		J12C XEM–FACING MC2 / 80			
A	MC2_BRK_80_21 MC2_BRK_80_22 MC2_BRK_80_23 MC2_BRK_80_24 MC2_BRK_80_25 MC2_BRK_80_26 MC2_BRK_80_27 MC2_BRK_80_28 MC2_BRK_80_29 MC2_BRK_80_30 MC2_BRK_80_31 MC2_BRK_80_32 MC2_BRK_80_33 MC2_BRK_80_34 MC2_BRK_80_35 MC2_BRK_80_36 MC2_BRK_80_37 MC2_BRK_80_38 MC2_BRK_80_39 MC2_BRK_80_40	21	PT_MC2_21	21	MC2_XEM_80_21 MC2_XEM_80_22 MC2_XEM_80_23 MC2_XEM_80_24 MC2_XEM_80_25 MC2_XEM_80_26 MC2_XEM_80_27 MC2_XEM_80_28 MC2_XEM_80_29 MC2_XEM_80_30 MC2_XEM_80_31 MC2_XEM_80_32 MC2_XEM_80_33 MC2_XEM_80_34 MC2_XEM_80_35 MC2_XEM_80_36 MC2_XEM_80_37 MC2_XEM_80_38 MC2_XEM_80_39 MC2_XEM_80_40	MC2_BRK_80_41 MC2_BRK_80_42 MC2_BRK_80_43 MC2_BRK_80_44 MC2_BRK_80_45 MC2_BRK_80_46 MC2_BRK_80_47 MC2_BRK_80_48 MC2_BRK_80_49 MC2_BRK_80_50 MC2_BRK_80_51 MC2_BRK_80_52 MC2_BRK_80_53 MC2_BRK_80_54 MC2_BRK_80_55 MC2_BRK_80_56 MC2_BRK_80_57 MC2_BRK_80_58 MC2_BRK_80_59 MC2_BRK_80_60	41	PT_MC2_41	41	MC2_XEM_80_41 MC2_XEM_80_42 MC2_XEM_80_43 MC2_XEM_80_44 MC2_XEM_80_45 MC2_XEM_80_46 MC2_XEM_80_47 MC2_XEM_80_48 MC2_XEM_80_49 MC2_XEM_80_50 MC2_XEM_80_51 MC2_XEM_80_52 MC2_XEM_80_53 MC2_XEM_80_54 MC2_XEM_80_55 MC2_XEM_80_56 MC2_XEM_80_57 MC2_XEM_80_58 MC2_XEM_80_59 MC2_XEM_80_60
		22	PT_MC2_22	22			42	PT_MC2_42	42	
		23	PT_MC2_23	23			43	PT_MC2_43	43	
		24	PT_MC2_24	24			44	PT_MC2_44	44	
		25	PT_MC2_25	25			45	PT_MC2_45	45	
		26	PT_MC2_26	26			46	PT_MC2_46	46	
		27	PT_MC2_27	27			47	PT_MC2_47	47	
		28	PT_MC2_28	28			48	PT_MC2_48	48	
		29	PT_MC2_29	29			49	PT_MC2_49	49	
		30	PT_MC2_30	30			50	PT_MC2_50	50	
		31	PT_MC2_31	31			51	PT_MC2_51	51	
		32	PT_MC2_32	32			52	PT_MC2_52	52	
		33	PT_MC2_33	33			53	PT_MC2_53	53	
		34	PT_MC2_34	34			54	PT_MC2_54	54	
		35	PT_MC2_35	35			55	PT_MC2_55	55	
		36	PT_MC2_36	36			56	PT_MC2_56	56	
		37	PT_MC2_37	37			57	PT_MC2_57	57	
		38	PT_MC2_38	38			58	PT_MC2_58	58	
		39	PT_MC2_39	39			59	PT_MC2_59	59	
		40	PT_MC2_40	40			60	PT_MC2_60	60	
B	J13D BRK–FACING MC2 / 80		J12D XEM–FACING MC2 / 80							
	MC2_BRK_80_61 MC2_BRK_80_62 MC2_BRK_80_63 MC2_BRK_80_64 MC2_BRK_80_65 MC2_BRK_80_66 MC2_BRK_80_67 MC2_BRK_80_68 MC2_BRK_80_69 MC2_BRK_80_70 MC2_BRK_80_71 MC2_BRK_80_72 MC2_BRK_80_73 MC2_BRK_80_74 MC2_BRK_80_75 MC2_BRK_80_76 MC2_BRK_80_77 MC2_BRK_80_78 MC2_BRK_80_79 MC2_BRK_80_80	61	PT_MC2_61	61	MC2_XEM_80_61 MC2_XEM_80_62 MC2_XEM_80_63 MC2_XEM_80_64 MC2_XEM_80_65 MC2_XEM_80_66 MC2_XEM_80_67 MC2_XEM_80_68 MC2_XEM_80_69 MC2_XEM_80_70 MC2_XEM_80_71 MC2_XEM_80_72 MC2_XEM_80_73 MC2_XEM_80_74 MC2_XEM_80_75 MC2_XEM_80_76 MC2_XEM_80_77 MC2_XEM_80_78 MC2_XEM_80_79 MC2_XEM_80_80					
		62	PT_MC2_62	62						
		63	PT_MC2_63	63						
		64	PT_MC2_64	64						
		65	PT_MC2_65	65						
		66	PT_MC2_66	66						
		67	PT_MC2_67	67						
		68	PT_MC2_68	68						
		69	PT_MC2_69	69						
		70	PT_MC2_70	70						
		71	PT_MC2_71	71						
		72	PT_MC2_72	72						
		73	PT_MC2_73	73						
		74	PT_MC2_74	74						
		75	PT_MC2_75	75						
		76	PT_MC2_76	76						
		77	PT_MC2_77	77						
		78	PT_MC2_78	78						
		79	PT_MC2_79	79						
		80	PT_MC2_80	80						
C										
D										
E										
F										
	NOT FOR FABRICATION: logical study; missing physical/protection design									
	Sheet: /08 / MC2 21–80 PASS–THROUGH/ File: MC2_PassThrough.kicad_sch									
	Title: MC2 contacts 21–80									
	Size: A3				Date: 2026–09–29			Rev: R8 CANDIDATE		
	KiCad E.D.A. 10.0.6							Id: 8/8		
	1	2	3	4	5	6	7	8		

All contact identities must be checked against Opal Kelly BRK Altium and XEM mating drawing before fabrication.